

$$= -42 + \cancel{102} 20$$

$$= 28$$

$$\cancel{28}$$

• $|A| \neq 0$ रत्न Non-singular matrix

$|A| = 0$ रत्न singular matrix

$$A^{-1} = \frac{1}{|A|} \cdot \text{Adj of } A$$

$$(*) \quad A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore |A| = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= 1 \cdot \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = 1 - 0 = 1 \neq 0$$

$$a_{11} = \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = 1 - 0 = 1$$

$$a_{12} = - \begin{vmatrix} 0 & 2 \\ 0 & 1 \end{vmatrix} = -(0 - 0) = 0$$

$$a_{13} = \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} = 0 - 0 = 0$$

$$a_{21} = - \begin{vmatrix} 2 & 3 \\ 0 & 1 \end{vmatrix} = -(2 - 0) = -2$$

$$a_{22} = \begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} = 1 - 0 = 1$$

Transpose Matrix:

- Matrix - U row & column are position change row - into

Ex: $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 4 & 5 & 3 \end{bmatrix}$

$\therefore A^T = \begin{bmatrix} 1 & 0 & 4 \\ 2 & 1 & 5 \\ 3 & 2 & 3 \end{bmatrix}$

Idempotent Matrix:

- $A^2 = A$

$$A = (A^T)$$

~~Idempotent~~

Nipotent Matrix:

- $A^k = 0 = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}$

$$A = A^T$$

- $k = 0, 1, 2, 3, \dots$

- In matrix $0^0 = 0$

$$A = A^T$$

Hermitian Matrix:

- A is square & complex matrix

Ex:

$$A = \begin{bmatrix} 2 & 3+i \\ 3-i & 5 \end{bmatrix}$$

$$\therefore A^T = \begin{bmatrix} 2 & 3-i \\ 3+i & 5 \end{bmatrix}$$

complex
conjugate
sign

$$\therefore (A^T)'$$

$$= \begin{bmatrix} 2 & 3+i \\ 3-i & 5 \end{bmatrix} = A$$

• Hermitian matrix $\{(A^T)' = A\}$

☐ Skew Hermitian Matrix:

$$\cdot (A^T)' = -A$$

☐ Symmetric Matrix:

$$\cdot A^T = A \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0 = "A"$$

☐ Skew symmetric Matrix:

$$\cdot A^T = -A$$

☐ Minor:

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 4 & 5 & 1 \\ 2 & 4 & 5 \end{bmatrix}$$

$$a_{23} = - \begin{vmatrix} 1 & 2 \\ 0 & 0 \end{vmatrix} = -(0-0) = 0$$

$$a_{31} = \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} = 4 - 3 = 1$$

$$a_{32} = - \begin{vmatrix} 1 & 3 \\ 0 & 2 \end{vmatrix} = -(2-0) = -2$$

$$a_{33} = \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = 1 - 0 = 1$$

$$\therefore \text{Cofactor matrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$$

$$\therefore \text{Adj of } A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{bmatrix}^T$$

$$= \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{1} \times \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{Ans})$$

HW:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix};$$

find $A^{-1} = ?$